MILITARY SPECIFICATION

POWER SUPPLY, ELECTRODIAGNOSTIC INSTRUMENTS 110 volt, 60 cycle, AC

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers power supply units, suitable for use as variable voltage sources for electrodiagnostic instruments (see 6.3).

1.2 Classification. Power supply units shall be of the following types, as specified for the appropriate stock number and item identification:

Type I - 2-circuit.
Type II - 1-circuit.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, or requests for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

TT-C-490 Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings.
TT-E-489 Enamel, Alkyd, Gloss (for Exterior and Interior Surfaces).
PPP-B-585 Boxes, Wood, Wirebound.
PPP-B-601 Boxes, Wood, Cleated-Plywood.
PPP-B-621 Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 Box, Fiberboard.
PPP-T-76 Tape, Pressure-Sensitive Adhesive Paper (for Carton Sealing).

Military

MIL-L-10547 Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof Flexible.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:12Z
Check the source to verify that this is the current version before use.

STANDARDS

Federal

Fed. Std. No. 595 Colors.

Military

MIL-STD-105 Sampling Procedures and Tables, for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

DRAWINGS

Defense Medical Supply Center

20092 Power Supply, Electrodiagnostic Instruments, 2 Circuit, 110 Volt, 60 Cycle, AC.
20093 Power Supply, Electrodiagnostic Instruments, 1 Circuit, 110 Volt, 60 Cycle, AC.

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNDERWRITERS' LABORATORIES, INC.

Underwriters' Laboratories Standard.

UL 498 Attachment Plugs and Receptacles.

(Underwriters' Laboratories Standards are available from the Underwriters' Laboratories, Inc., 161 Sixth Avenue, New York 13, N.Y., 207 East Ohio Street, Chicago 11, Ill., and 1655 Scott Blvd., Santa Clara, Calif.)

3. REQUIREMENTS

3.1 Materials. Satisfactory high-grade materials shall be used throughout. Sheet steel entering into the construction of the power supply units shall be in strict accordance with the applicable drawings and bills of materials. The materials shall be entirely satisfactory for the purpose intended and of sufficient strength to withstand the shock of normal usage and to maintain operational alignment for the life of the items.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:12Z
Check the source to verify that this is the current version before use.

3.2 Construction and design. Type I (2-circuit) and type II (1-circuit) power supplies shall be in strict accordance with Defense Medical Supply Center Drawings 20092 and 20093. The type I power supply shall weigh no more than 9-1/2 pounds, and the type II power supply shall weigh no more than 5-1/2 pounds. Electrical components shall be mounted in a steel enameled case provided with a carrying handle. The type I power supply shall provide two independent circuits, each capable of providing an alternating-current variable output voltage source from 0 to 6 volts (tolerance: plus 0.5 volt, minus 1.0 volt), and a minimum output current of 0.85 ampere. The electrical output requirements for the type II power supply shall be the same as for the type I, except that there shall be only a single circuit. Each circuit of the power supplies, when tested as specified in 4.4.1 and 4.4.2, shall meet the electrical requirements indicated above. The unit shall pass the dielectric strength test specified in 4.4.4.

3.2.1 Circuit breakers. The 2-circuit unit shall provide two 0.1-ampere ± 10 percent push-to-reset circuit breakers to be employed in each of the 110-volt primary windings. The 1-circuit unit shall provide one 0.1-ampere ± 10 percent push-to-reset circuit breaker to be employed in the 110-volt primary winding for protection. Circuit breakers shall pass the test specified in 4.4.3.

3.2.2 Transformers. Two transformers for type I units, and one transformer for type II units, shall be provided for the dual function of reducing the voltage from 110 to 6 volts, alternating current and isolating the patient from the power source. The transformers shall be hermetically sealed with a breakdown voltage of at least 1500 volts root mean square and a secondary rating of 2 amperes at 6.3 volts.

3.2.3 Service cable. The types I and II units shall be provided with a service cable, as indicated on the applicable drawing. The cable shield shall be grounded to the instrument case, and fitted at the opposite end with a 3-prong, grounded-type plug. Plug shall conform to Underwriters' Laboratories Standard UL 498. An adapter shall be provided for use at installations equipped with 2-prong receptacles.

3.2.4 Indicator lights. White indicator lights with identifying nameplates or screened markings shall be provided to indicate either "open" or "short" circuit conditions. Red indicator lights shall indicate that the power is "on" and shall operate in conjunction with the double-pole "off-on" switches.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:12Z
Check the source to verify that this is the current version before use.

3.3 Interchangeability of parts. All individual parts or components of any complete unit shall be interchangeable with the corresponding parts or components of all other complete units.

3.4 Finish. All outside surfaces of the power supplies shall be finished with a hard and smooth baked enamel, free of wrinkles, grit, and bare areas. All inside surfaces of the case, stand, and brackets shall be treated by the method outlined in 3.4.2 but need not be primed and painted.

3.4.1 Color. The color shall be gray, conforming to color 16473 (see 6.2) of Federal Standard 595.

3.4.2 Surface treatment. Surfaces to be painted shall first be thoroughly cleaned as specified for type I of Federal Specification TT-C-490. All welds shall be thoroughly and completely cleaned of flux, scale, etc., prior to the application of the primer.

3.4.3 Primer. After cleaning, one coat of suitable commercial primer shall be applied. The primer shall be applied so as to produce a uniform surface free of pebbles and wrinkles.

3.4.4 Enamel. After the primer has been applied, two coats of synthetic gloss enamel, conforming to class B of Federal Specification TT-E-489, shall be applied and baked on in accordance with the temperature schedule recommended by the paint manufacturer. The enamel shall produce a final, uniform, and smooth surface without runs or wrinkles. The total dry-film thickness shall not be less than 0.003 inch at any point.

3.5 Identification marking. Each unit shall be legibly and permanently marked with the stock number, item identification, name or registered trademark of the manufacturer, contract or purchase order number, electrical data and the letters "U.S.". "The "U.S." marking shall be applied in a conspicuous area on the back of the case, apart from the balance of the identification marking in letters at least 1/4-inch in size.

3.6 Service data. Two copies of one or more booklets containing complete instructions for installation, operation, maintenance, trouble shooting and a list of component parts shall be enclosed with each unit. Wiring diagrams shall be included for electrical items or components. Each part of the unit other than common fasteners, etc., shall be identified in the service data and parts list by the manufacturer's part number. As soon as practicable after award of the contract, the contractor shall furnish to the Contracting Officer, via the cognizant Government inspector, two preliminary copies of his proposed service data, for review and comment by the Defense Medical Supply Center. The preliminary service data shall be submitted sufficiently early to permit adequate review by the Defense Medical Supply Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of required revisions and corrections in the final service data without delaying delivery under the contract. In addition, the contractor shall mail 15 copies of the approved service data to the contracting officer for distribution.

Source: https://assist.dla.mil -- Downloaded: 202610-21T08:12Z
Check the source to verify that this is the current version before use.

3.7 Workmanship. Workmanship shall be first class throughout. The power supply units shall be free from defects which detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with Military Standard MIL-STD-105.

4.2.2 For tests. Sampling for tests shall be conducted in accordance with Military Standard MIL-STD-105.

Characteristic	Requirement	Test procedure	Inspection level	AQL % defective
Open circuit	3.2	4.4.1	S3	2.5
Short circuit	3.2	4.4.2	S3	2.5
Circuit breaker	3.2.1	4.4.3	S3	2.5
Dielectric strength	3.2	4.4.4	100%	--

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:12Z
Check the source to verify that this is the current version before use.

4.3 Examination. The power supply units shall be examined to determine compliance with all requirements contained in this specification.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Open circuit. With the rheostat set at the maximum setting, and the output open circuited, the voltage across the output terminals shall be within the tolerances specified in 3.2.

4.4.2 Short circuit. With the rheostat set at the maximum setting, the output short circuited, and with an ammeter inserted in series with the output, the current through the ammeter shall not be less than that specified in 3.2.

4.4.3 Circuit breaker. A variable resistance shall be placed across the secondary of the power supply transformer and an ammeter inserted in series in the primary circuit. This resistance shall be reduced until the circuit breaker in the primary circuit operates to cut the flow of current. The current, as indicated on the ammeter, at the point where the circuit breaker opens the primary circuit, shall be recorded for comparison with the requirement (including tolerance) specified in 3.2.1.

4.4.4 Dielectric strength. A 60-cycle 900-volt potential shall be developed between the live-metal parts of the primary circuit and the dead-metal parts of the power supply unit. Starting at zero, the applied potential shall be increased gradually until the required voltage has been reached, and shall be held at that value for 1 minute. The test shall be repeated between the live-metal parts of the secondary cirucit and the dead-metal parts of the power supply unit. The test shall also be performed between the live-metal parts of the primary circuit and the live-metal parts of the secondary circuit, except that the potential shall be increased to 1500 volts. Care shall be taken to insure that all switches and circuit breakers are in the closed position during the test, so that the test voltage is applied to all live-metal parts.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One (1) power supply, as specified, constitutes one unit of issue.

Source: https://assist.dla.mil -- Downloaded: 202610-21T08:12Z
Check the source to verify that this is the currentversion before use.

5.1.2 Packaging quantities. The number of units of issue indicated in table I shall be packaged in each unit and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table I - Packaging quantities

Type	Unit package	Intermediate package	Exterior container
Type 1	1 unit	Not required	6 units
Type 2	1 unit	Not required	12 units

5.1.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Each unit shall be packaged in a carton of appropriate size constructed in accordance with Federal Specification PPP-B-636. Each carton shall be adequately secured to prevent accidental opening.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with Federal Specification PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type III, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type I, class 2.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to Military Specification MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

Source: https://assist.dla.mil -- Downloaded: 20261O-21T08:12Z
Check the source to verify that this is the currentversion before use.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix of Federal Specification PPP-B-636. In addition, all seams, including manufacturer's joint, shall be closed by tape conforming to the requirements of Federal Specification PPP-T-76. The tape shall be not less than 2 inches wide. Three strips shall be applied to both top and bottom so that all seams and joints shall be covered their full length. Tape applied over the lengthwise seams shall extend a minimum of 2 inches onto each of the end panels. Tape applied over the end edge seams shall extend a minimum of 2 inches along the side and top, and side and bottom panels.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall conform to Federal Specification PPP-B-601, table I, using type III, class 1 plywood; or shall be designed for type 2 load, and constructed in accordance with Federal Specification PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-636, type I, class 1.

5.3 Marking.

5.3.1 Interior packages. Each unit package shall be marked in accordance with Military Standard MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with Military Standard MIL-STD-129. In addition, the letter "L" shall precede the Federal stock number. One space shall be allowed between the letter "L" and the first digit of the stock number. The terms "Stock number" or "FSN" shall not be indicated.

6. NOTES

6.1 Ordering data. Procurement documents, should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:12Z
Check the source to verify that this is the current version before use.

6.2 Availability of color chips. The individual 3- by 5-inch chips of the color referenced in 3.3.1 are available from the General Services Administration, Business Service Center, Region 3, Seventh and D Streets, S.W., Washington 25, D. C. Price 5 cents.

6.3 This specification covers power supply units, suitable for use as variable voltage sources for the following electrodiagnostic instruments:

Stock No.	Item Identification
6515-323-7400	CYSTOSCOPE, Modified Young, Convertible, Double Catheterizing, 21 or 22 Fr
6515-323-7800	CYSTOSCOPE, Modified Young, Convertible, Double Catheterizing, 24 Fr
6515-323-8800	CYSTOSCOPE, Modified Young, Double Catheterizing, 16 Fr
6515-324-2200	PAN-ENDOSCOPE
6515-340-8100	LIGHT, HEAD, DIAGNOSTIC, 3.8 volt, AC-DC
6515-357-9510	SIGMOIDOSCOPE, Electric, Distal and Proximal Lighting, 63 Fr
6515-359-8000	RESECTOSCOPE, PROSTATIC, 28 Fr
6515-360-6100	RETRACTOR SET, BRAIN, Frazier, Set of 3
6515-754-0165	CYSTOSCOPE, Brown Buerger, Convertible, Double Catheterizing, 21 Fr
6515-754-0169	CYSTOSCOPE, Brown Buerger, Convertible, Double Catheterizing, 24 Fr

Source: https://assist.dla.mil -- Downloaded: 202610-21T08:12Z
Check the source to verify that this is the current version before use.

6.4 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Military requirements.

Custodians:
Army. - MD
Navy. - BuMed
Air Force. - AFMSG

Preparing activity:
Defense Supply Agency. - DM

Review activities:
Army. - MD
Navy. - BuMed
Air Force. - AFMSG

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:12Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

FORM APPROVED BUDGET BUREAU NO. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use of the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION	
MIL-P-36196(DM), 4 May 1964	POWER SUPPLY, ELECTRODIAGNOSTIC INSTRUMENTS, 110 volt, 60 cycle, AC

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS (ATTACH ANY PERTINENT DATA WHICH MAY BE OF USE IN IMPROVING THIS SPECIFICATION. IF THERE ARE ADDITIONAL PAPERS, ATTACH TO FORM AND PLACE BOTH IN AN ENVELOPE ADDRESSED TO PREPARING ACTIVITY)

SUBMITTED BY (PRINTED OR TYPED NAME AND ACTIVITY)	DATE

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:12Z
Check the source to verify that this is the current version before use.

FOLD

DEFENSE SUPPLY AGENCY
DEFENSE MEDICAL SUPPLY CENTER
3RD AVENUE AND 29TH STREET
BROOKLYN, N.Y., 11232

OFFICIAL BUSINESS

POSTAGE FEES PAID
DEFENSE SUPPLY AGENCY

Defense Supply Agency
Defense Medical Supply Center
ATTN: Directorate of Technical Operations (Code T-M2)
3rd Avenue and 29th Street
Brooklyn, N.Y., 11232

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:12Z
Check the source to verify that this is the current version before use.